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RACE, OPPORTUNITY, AND DIVERSITY OF SOCIAL CIRCLES IN MANAGERIAL NETWORKS

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This study investigated the informal networks of white and minority managers. Minority managers had more racially heterogeneous and fewer intimate network relationships. Within the minority group, differences in advancement potential were associated with different network configurations: high-potential individuals balanced same- and cross-race contacts; others had networks dominated by ties to whites. High-potential minorities also had more contacts outside their groups, fewer high-status ties, and less overlap between their social and instrumental circles. Relative to whites, minority managers viewed similar network characteristics as providing less access to career benefits.

A central theme that has emerged in the organizational literature on racial minorities in corporate settings is the difficulties members of minority groups experience gaining significant social and instrumental support in the workplace (Thomas & Alderfer, 1989). Many have argued that exclusion from social networks explains the failure of minority managers to advance more rapidly in their careers and organizations (Dickens & Dickens, 1982; DiTomaso, Thompson, & Blake, 1988; Fernandez, 1991; Irons & Moore, 1985; Morrison & Von Glinow, 1990), yet network differences by race are rarely examined empirically. Further, the scant existing literature offers contradictory recommendations: an assimilation paradigm suggests that members of racial minorities should develop interpersonal networks that are similar to those of their successful white peers; in contrast, a pluralist perspective implies that because of different circumstances, minority members must use different approaches to achieve similar ends (Cox, 1991; Nkomo, 1992; Thomas, 1993). Given the importance of informal networks for career success (Brass, 1985; Burt, 1992), investigating them may shed light on the nature of the obstacles racial minorities face in the workplace and strategies for overcoming those obstacles.

The reported study drew from social network and race relations theories to illuminate relationships among race, career outcomes, and network

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interaction patterns. My central thesis was twofold: the racial composition of a managerial network is highly constrained by the macrocontext in which interpersonal interaction is embedded, and decisions about network composition constitute the central strategic choice made at the individual level. Macrocontext, or the pattern of different racial groups' representation in organizational and societal power elites, produces systematic differences in network structure that factors such as "human capital" investment or position power cannot explain. Microlevel factors, however, particularly variables that shape individuals' goals and network development strategies (Wharton, 1992), also produce network differences. To investigate the role of these macro and micro factors and the relative explanatory values of the assimilation and pluralist perspectives, I compared the structures of the informal networks of white and minority middle managers and explored explanations for intraminority group variance in network patterns.

Recent research (Burt, 1992) further suggests that exclusive focus on the presence or absence of network differences may be misleading since similar networks may not necessarily provide similar benefits to members of different groups. Since individuals' views about what constitutes a useful network may affect their network strategies, the match between network structure and perceptions of network utility warrants exploration. The discovery of conditional effects would provide additional evidence of network development trade-offs that result from macrostructural factors and are unique to minorities. Thus, I also investigated the extent to which the relationship between network characteristics and managers' assessments of the utility of their networks varied by race.

EFFECTS OF RACE ON MANAGERIAL NETWORKS

Networks as Social Resources

An informal managerial network is defined here as the set of job-related contacts that a manager relies on for access to task-related, career, and social support. In the sociological literature, a network's instrumental utility is viewed as a function of two dimensions: the network's range, which refers to the diversity of group affiliations encompassed and the potential access to information and resources from diverse and distant subgroups afforded by the network (Burt, 1992; Campbell, Marsden, & Hurlbert, 1986; Granovetter, 1973), and its status, which refers to the positions of network contacts in the relevant status hierarchy (Lin, 1982). Range provides both useful information and bargaining opportunities based on individuals' control over their contacts. Thus, people whose network contacts extend beyond their required work flow interactions and immediate work groups or units tend to be more powerful (Blau & Alba, 1982; Brass, 1984). Reaching diverse others, however, is not sufficient if few contacts are high enough in status to be instrumentally useful (Campbell et al., 1986; Lin, 1982); access to peers, superiors, and an organization's "dominant

coalition" are critical for power and advancement (Brass, 1984; Kotter, 1982).

The organizational literature has additionally emphasized the importance of close informal bonds of trust and loyalty that ensure reliability under conditions of uncertainty (Kanter, 1977). Although range and status pertain to the probable instrumental value of what may be exchanged, the closeness of a relationship determines the propensity of the contact to transmit a benefit to a particular individual and not to another (Burt, 1992). From a career development perspective, close relationships are more likely to fulfill psychosocial functions (Kram, 1988). Psychosocial functions are aspects of a relationship that enhance an individual's sense of competence, identity, and effectiveness in a professional role; they include serving as a role model, acceptance, and friendship. Psychosocial functions are distinguished from purely instrumental functions, such as providing exposure to senior management and advocacy for promotion, in that they involve benefits that stem from the nature of the relationship rather than the positional power of the contact.

Multiplexity is defined as the degree to which network relationships are multidimensional or, at an aggregate level, the extent to which network circles formed for different purposes overlap (have common members; Granovetter, 1973). The classic manifestation of a multiplex tie is socializing outside of work between co-workers; multiplex relationships are also associated with high trust and reliability since both parties have had the opportunity to interact and get to know each other in a variety of contexts. The career benefits of a particular type of strong, multiplex tie, the mentor-protégé relationship, are well documented (Kram, 1988; Thomas, 1990).

A network characteristic that tends to be overlooked in discussions of networks as social resources is homophily, defined as the degree to which pairs of individuals who interact are similar in identity or organizational group affiliations (Marsden, 1988; Rogers & Kincaid, 1981). Its relevance to the study of access to instrumental resources derives from findings that interpersonal similarity increases ease of communication, improves predictability of behavior, and fosters relationships of trust and reciprocity (Kanter, 1977; Lincoln & Miller, 1979). Consequently, explanations of network obstacles must consider the consequences of a defining feature of corporate life for minorities: the scant availability of similar—same-race—others for informal contact (Ibarra, 1993).

Macrostructural Constraints on Network Structure

A central tenet of social network theory is that any set of social relationships is embedded within a larger structural context that precludes or makes possible various kinds of social contacts; organizational demography, intergroup relations, and the distribution of valued resources within a social system constrain the development of informal relationships between individuals (Alderfer, 1986; Blau, 1977). In the United States,

African Americans, Hispanics, and Asian Americans are underrepresented in private-sector managerial positions, particularly at the highest levels (Cox, 1991; Fernandez, 1991; Morrison & Von Glinow, 1990). The representation of minorities in the ranks of managers in the *Fortune* 500, for example, averages less than 12 percent, and gaps of 15 to 30 percentage points commonly exist between the proportions of minority members in the overall managerial ranks and in middle and higher levels of management in those companies (Cox, 1991).

Workplace information interaction, in turn, is embedded in a broader structure of social relations that extends beyond the boundaries of a firm. By virtue of societal patterns such as institutional segregation, members of minority groups often develop relationships within and across nonoverlapping social circles. The concept of biculturalism has been used to explain the experience of African Americans, who require access to both the black community and the dominant culture in order to obtain social support and job-related resources (Bell, 1990; Thomas & Alderfer, 1989). The immediate consequences of such a structural context for a managerial network are threefold. First, minority managers have a much smaller set of same-race others with whom to have informal interaction (Marsden, 1988). In contrast to white managers, minority managers will have the majority of their informal encounters with dissimilar others.

Hypothesis 1: Minority managers will have a smaller proportion of homophilous (same-race) ties in their personal networks than will their white counterparts.

Second, homophily strengthens interpersonal bonds (Kanter, 1977; Lincoln & Miller, 1979). An important component of tie strength is intimacy, that is, the interpersonal closeness of a network relationship (Granovetter, 1973). Cross-race relationships tend to be weaker than same-race ties (Thomas, 1990), and informal social relations (friendship ties) tend to develop between people who share commonalities, including race and gender (Ibarra, 1992; Lincoln & Miller, 1979; Tsui & O'Reilly, 1989). Thomas (1990) found that racial difference was often an obstacle for white mentors in identifying positively with their African American protégés, and Tsui and O'Reilly (1989) reported that demographic similarity affected superiors' personal attraction to and identification with subordinates. In research on peer relations, South, Bonjean, Markham, and Corder (1982) found that homophilous relationships provided individuals with greater social support from co-workers than did heterophilous relationships.

Hypothesis 2: Minority managers will have a smaller proportion of intimate ties in their personal networks than will their white counterparts.

Third, weaker cross-race relationships, coupled with nonoverlapping social circles outside the workplace, may produce nonoverlapping social and work-related circles in the workplace. Qualitative reports have suggested that racial minorities feel excluded from a variety of social activi-

ties that take place after working hours (e.g., Davis & Watson, 1982; Fernandez, 1991); demographically different individuals have also been found to be the least socially integrated within their groups (Kanter, 1977; O'Reilly, Caldwell, & Barnett, 1989). Further, minority group members often manage two different social circles in the workplace, developing both minority networks, which provide social support and information or advice on unique issues faced by minorities, and ties to the majority group, which provide access to important resources (Dickens & Dickens, 1982; Thomas, 1990). The net structural effects are race differences in multiplexity (the degree to which managerial networks include distinct circles that serve different purposes).

Hypothesis 3: Minority managers will have fewer multiplex network ties than their white counterparts.

As discussed in more detail below, differences in the range and status of contacts are not hypothesized to arise directly from macrostructural constraints; rather, they are expected to vary as a function of purposive network development strategies.

Microstrategies: Homophily, Career Success, and Identity

Despite the presence of significant structural constraints, people play an active role in structuring their social networks to achieve their goals. Their implicit or explicit strategies are based on theories of action, or "sets of guiding assumptions about situated orientations toward one another and the world" (DiMaggio, 1992: 118). Thomas (1993) argued that one important set of assumptions concerns appropriate and preferred ways of addressing race-related matters in the workplace. This article's argument is that for minority managers, such assumptions are reflected in the extent to which they develop informal ties to other minority group members, a central strategic choice in network development. Two different theoretical perspectives offer contrasting hypotheses about homophily within a minority group.

The "deficit hypothesis" posits that minority managers tend to have less instrumentally useful networks because they are less politically savvy in the ways of the white corporate world (Nkomo, 1992). This hypothesis implies assimilation, defined as "a unilateral process by which minority culture members adopt the norms and values of the dominant group in the organization" (Cox, 1991: 35). For network development, it suggests that if racial minorities developed networks structured like those of their successful white counterparts, they would reap similar instrumental benefits. In particular, given the distribution of power and resources within most organizations and in society at large, an informal network that is predominantly composed of majority group contacts will be most beneficial for instrumental purposes (Lin, 1982). This would be true for both functional and symbolic reasons: Majority group members have greater access to power and resources through their formal positions and informal net-

work connections, and informal contact with the majority group may be viewed as evidence of assimilation and conformity to corporate norms and culture. The implication for the present study is that minority managers who have been assessed as having greater potential for further advancement would be expected to evidence more network contact with the majority group than less mobile minority group members.

The alternative perspective, advocated in this study, is based on theories of pluralism, racial identity, and intergroup relations. A pluralist perspective suggests that for a minority manager, a critical task is developing a positive sense of racial identity while at the same time adopting those aspects of the dominant culture necessary to be effective (Fernandez, 1990; Thomas, 1993). Racial identity theory (Davis & Watson, 1982; Dickens & Dickens, 1982) posits that the process of coming to grips with one's racial identity follows several phases, from efforts to be accepted within the dominant culture, to a rejection or avoidance of contact with the dominant culture beyond that necessary for "getting the job done," to a final stage in which the individual develops a style that is consistent with corporate norms but maintains his or her own uniqueness or sense of self. Each stage represents a dominant stance toward one's own racial group and the dominant group; each has a behavioral component: the individual's seeking to develop relationships primarily within the dominant culture, primarily within the minority group, or with people in both groups. Finally, according to intergroup theory (Alderfer, 1986), racial division and domination in organizations and society require minorities to use means that differ from those that are effective for majority group members to attain similar results. Following recent reviews of the empirical literature (Cox & Nkomo, 1990a), I develop a pluralist perspective on the relationship between network structure and career outcomes.

With regard to homophily, a pluralist perspective suggests that work-related informal contact across and within majority and minority groups is critical for career success. Minority managers provide each other with mutual support. But the role of these homophilous ties is not exclusively expressive; instrumental benefits include learning from other minority group members about interpersonal and behavioral strategies, such as strategies for managing racial barriers (Davis & Watson, 1982; Dickens & Dickens, 1982; Thomas, 1990, 1993). More generally, minority contacts function as necessary channels for accurate information, cooperation in getting things done, and career development functions, including role-modeling. Cross-race contacts, on the other hand, are necessary for functioning effectively in a white-dominated context and also provide a variety of instrumental and psychosocial supports (Thomas, 1993). Thus, this stream of research and theorizing suggests that minorities with greater potential for advancement will develop a mix of same-race and cross-race relationships, in contrast to both their white and less mobile minority counterparts, who are expected to have networks dominated by ties within the majority group.

Hypothesis 4: Minority group members with high potential for organizational advancement will have a higher proportion of minority contacts in their informal networks than both their white counterparts and less mobile minority counterparts.

Cox and Nkomo (1990b) suggested that since minorities are still not accepted in corporate inner circles, mainstream strategies for networking may backfire for them. The arguments above suggest that homophily will be the principal indicator of alternative, nonassimilationist network strategies; however, organizational characteristics of a network, such as its range and status, also indicate strategies that are expected to differentiate high-potential minority group members from their white peers and less mobile minority counterparts. This argument is based on two notions. The first, directly related to homophily, reflects the fact that macrostructure constrains microstrategies: Since the number of minority managers is low in most organizations, a network high in its proportion of minorities will differ in other ways from a set of contacts predominantly drawn from the majority group. Second, as discussed below, homophily is only one indicator of a broader underlying strategy, one that emphasizes access to a diversity of social circles, defined on the basis of both identity and organizational group memberships.

With respect to network range, if racial minorities seek ties to members of their identity groups, they are likely to have to reach out further in their organizations, beyond their immediate peers and superiors or functional areas. Thomas (1990), for example, found that black managers had more extradepartmental relationships and broader support networks than whites, as a result of their relationships with other blacks who were located outside their departments. Although partially a product of homophily, a strategy of developing a high proportion of extradepartmental contacts may indicate a cosmopolitan orientation that also leads to the development of other types of extragroup relationships. Beyond the benefits of access to same-race others, widely ranging networks may contribute more generally to career success by providing links to alternative positions and opportunities (Burt, 1992; Granovetter, 1982; Lin, 1982). Cross-boundary mobility may be more important for minority group members than for whites, since the former are more likely to encounter racial barriers (Morrison & Von Glinow, 1990).

Hypothesis 5: Minority group members with high potential for organizational advancement will have networks with wider range than both their white counterparts and less mobile minority counterparts.

Second, if minorities are disproportionately located in the lower echelons of most corporations, a network with a large proportion of minority members will be lower in status than a network with a different de-

mographic composition (Thomas, 1990). Thus, although status is generally expected to bear a positive association to career success (see Burt [1992] for a review), the low number of minorities in top decision-making positions in predominantly white corporations means that, for minority managers, the benefits of having high-status contacts are likely to be in direct competition with the psychosocial benefits that are more easily obtained in interaction with like others (Thomas, 1993). Alternatives include peer networks, which also serve as sources of psychosocial support (Kram & Isabella, 1985). Again, membership in a network whose members have lower status implies that an individual has made a strategic trade-off to gain a balance of instrumental and psychosocial benefits and emphasizes using a diversity of groups as social resources.

Hypothesis 6: Minority group members with high potential for organizational advancement will have lower-status network contacts than both their white counterparts and less mobile minority counterparts.

Network Structure and Perceived Access to Network Resources

The discussion above implies that similar network characteristics may not provide access to the same benefits for members of different identity groups. For example, if racial dissimilarity means that relationships with people of higher rank are unlikely to provide sufficient psychosocial support (Thomas, 1993), minority managers may not view high status as being as useful as a white manager would view it. Similarly, observers may not view a broad network that provides access to other minorities as providing the necessary internal information and sponsorship within a manager's subunit. In brief, because minority managers face trade-offs that their white counterparts do not confront, network characteristics that are associated with career success may not necessarily lead the minority members to view their networks as useful or adequate. Perceptions of what makes for a useful network are important because they are likely to affect network strategies for balancing competing network requirements.

Consistent with the pluralist perspective outlined above, my expectation was that the relationship between network characteristics and perceptions of having a useful network would differ, by race, in three ways. First, although a network that is broad in range may be necessary for access to instrumental resources, broad networks may not be as useful to individuals from groups lacking in "systemic legitimacy" (Burt, 1992; Granovetter, 1982). As Burt argued, when the legitimacy of an individual's position in a power structure is implicitly questioned, as is often the case for minorities, the information and brokerage benefits of broad networks become less important than the necessity of becoming a "known quantity," a goal often achieved through local sponsorship. Assuming that an indi-

vidual has limited time and energy for network development, attenuation of the time available for building within-group support may offset the advantages of a widely ranging network. This time effect is exacerbated when a lack of homophily poses barriers to the development of trust and attachment (Thomas, 1993; Thomas & Alderfer, 1989). Thus, for reasons other than the positive association between network range and career success for minorities discussed above, white managers with broad networks may be expected to view their networks as more instrumentally useful than minority managers with broad networks.

Hypothesis 7: Having a broad range of network contacts will be more predictive of perceived network utility for white than for minority managers.

Second, to the extent that minority group members' relationships with superiors are the ties most likely to be cross-racial, the benefits of sharing a common background are forfeited. Further, network relationships between people who are doubly dissimilar are more difficult to develop (Kaplan, 1984; Lin, 1982): A relationship with a superior may be heterophilous with respect to both identity (race) and organizational (rank) groups for a minority manager (Alderfer, 1986), but for a white manager, it may only be dissimilar on one dimension. Homophilous ties may not, however, provide access to an organization's power structure. One way minority managers manage these competing requirements is to establish dual support structures. Thomas (1990) found, for example, that black managers often relied on peer mentors because senior blacks were unavailable in their firms. Thus, the relationship between the status of contacts and perceptions of having a useful network is expected to be stronger for white managers.

Hypothesis 8: Having high-status contacts will be more predictive of perceived network utility for white than for minority managers.

Third, although a high proportion of intimate relationships among network contacts may be detrimental for managers from traditional insider groups, such as white men, because it can curtail their autonomy, for those outside organizational inner circles, such as members of racial minorities, intimate relationships may help signal legitimacy in the power structure (Burt, 1992). This hypothesis is consistent with research indicating that stereotypes and attribution biases are most prevalent in the absence of information obtained through personal experience (Berger, Fisek, Norman, & Zelditch, 1977) and that there is greater potential for bias and stereotyping in superficial relationships (Allport, 1954). Consequently, intimate relationships are expected to have greater instrumental value for minority than for white managers. Multiplex ties should have a similar effect since having informal contact in multiple contexts may reduce the uncertainty associated with cross-racial interaction.

Hypothesis 9: Having a high proportion of intimate and multiplex ties will be more predictive of perceived network utility for minority than for white managers.

METHODS

Research Sites and Respondents

This research was conducted as part of a larger study of the informal networks of middle managers in four *Fortune* 500 firms. The four sites were divisions or business units of large, mature bureaucratic organizations operating in different industries (telecommunications and pharmaceutical, automotive, and photographic products) and geographical regions.

Respondents were 63 middle-level managers. Given previous research indicating that most identity-group differences in informal networks disappear when structural variables such as occupation and hierarchical rank are held constant (Ibarra, 1992; Moore, 1990), I chose individuals occupying equivalent middle management ranks as potential respondents. Since each company had a different grade-level structure, I tried to ensure comparability across companies by matching respondents on specific responsibilities and distance from the bonus-eligible, executive ranks of their divisions or business units.

A second selection criterion was demographic diversity. The group was selected to be balanced by race and gender rather than to reflect the demographic composition of the organizations. The final set of respondents included 46 white managers, 20 of whom were women, and 17 members of minority groups, 5 of whom were women. The minority group included 12 African Americans, 3 Hispanics, and 2 Asian Americans. Third, in order to include a proxy for advancement opportunity, I asked the participating firms to “oversample” managers who had been placed in a fast-track or high-potential program. The high-potential programs are a critical part of each firms’ formal succession planning processes: Supervisors and subunits’ human resource management staffs nominate managers for advancement to the executive ranks on the basis of performance and potential, making final selections in consultation with division executive committees. Membership in the high-potential group indicated both an individual’s perceived effectiveness to date and the likelihood of future advancement to a rank several levels higher than his or her present rank. The high-potential subgroup included 27 white and 9 minority managers.

Finally, although the group represented a mix of staff and line managerial jobs, care was taken to avoid selecting a group in which minorities were overrepresented in staff positions and whites in line jobs or in which non-high-potential managers were overrepresented in staff positions and high-potential managers held primarily line jobs (48 percent of non-high-potential managers and 47 percent of high-potential managers

held line jobs; 47 percent of the minorities and 48 percent of whites held line jobs).

The selection process was undertaken by a human resources representative at each site in consultation with me. Rather than asking the representatives to select random samples, I instructed them to locate high-potential minority managers first (again, without oversampling staff jobs) and then to match those individuals as closely as possible with high-potential whites and non-high-potential minorities in similar jobs. Although the demographic distribution of respondents was not representative of that of the participating firms, the company representatives assured me that the job profile of the respondents were similar to those of their middle management population.

My original plan was to include 12–18 managers from each of four research sites; the actual numbers were 18, 18, 11, and 16 from companies 1, 2, 3, and 4, respectively. Of these, the proportion of minority respondents from each firm was .28, .33, .33, and .25. Participation was voluntary; all managers who were identified agreed to participate in the study. The companies did not differ significantly with regard to the racial composition of their managerial ranks: Approximately 10–12 percent of each site's managers were members of racial minority groups, a figure consistent with current assessments of *Fortune* 500 demographics (Cox, 1991). Information on the proportion of African American managers (who composed the majority of the minority group in this study) was also available for three of the four companies; it was 9.9, 8.7, and 6.7 for companies 1, 2, and 3, respectively. Similar data were not available for company 4, but my observation was that the percentage was nearest that of company 3. Overall, there were no race-based differences between the minority and white managers in age (38.9 vs. 40.0, $t = -0.62$, n.s.), tenure (12.4 vs. 15.8, $t = -1.6$, n.s.), or type of job (47 vs. 48 percent in line jobs, $t = -0.05$, n.s.). A significant difference did exist with regard to education, with minority managers more likely than whites to hold graduate degrees (76 vs. 46 percent, $t = 2.23$, $p < .05$), but there were no race-based differences in having a master's of business administration degree ($t = 0.45$, n.s.)

Data Collection

Data collection involved administering two segments of a sociometric questionnaire and conducting an approximately one-hour-long interview based on a semistructured interview guide. Participants were assured that their responses would be kept confidential. The interview began with preliminary questions about respondents' current jobs, followed by administration of the first segment of the questionnaire, which consisted of five network, or "name generator," questions. These questions asked respondents to list their contacts in five domains: (1) information—people "who have been valuable sources of information for you in your current job"; (2) advice—people who are "important sources of professional ad-

vice, whom you approach if you have a work-related problem or when you want advice on a decision you have to make"; (3) friendship—people "whom you consider to be personal friends, that is, those people you see most frequently for informal social activities such as going out to lunch, dinner, drinks, visiting one another's homes, and so on"; (4) career—people "who you feel have contributed most to your professional growth and development. Please include people who have taken an active interest in and concerted action to advance your career"; and (5) cooperation—"people whose help, support or cooperation you have successfully enlisted towards the accomplishment of your objectives."¹

These responses were then used as a basis for open-ended interview questions about the respondents' network development strategies and approaches. (Appendix A provides the interview guide.) One set of questions—"How important is it to you to develop network relationships with other minorities? Do you seek out other minorities as network contacts?"—was asked exclusively of the minority managers. Since the interview tended to prompt recall of additional contacts, respondents were permitted to add names to their five lists throughout the interview, but not after they were given the second part of the sociometric questionnaire.

When the interview was completed, respondents filled out the second portion of the sociometric questionnaire, containing brief items with Likert-type responses and a background information sheet. The second portion of the network questionnaire asked respondents to provide demographic and organizational information about each contact: the person's gender, race, organizational affiliation, and hierarchical rank; the frequency of interaction and the intimacy of the relationship between focal individual and contact; and whether or not their contacts knew each other. This part of the questionnaire was modeled after the network segment of the General Social Survey (Burt, 1984) and provided the raw data used to measure all but the multiplexity network measures.

Measures

Network characteristics. Four indicators of the *range* of an individual's network, or the diversity of ties across organizational boundaries (Campbell et al., 1986), were computed: the proportion of total ties to people outside the manager's business unit or division, the proportion of total ties to people outside the company, the density of the network, and the average frequency of interaction. Network density was measured as the proportion of possible ties among a focal individual's network contacts

¹ In a review of research evidence, Marsden (1990) concluded that network indexes are largely reliable when measures are taken to facilitate individuals' capacity to recall and report their network links accurately. Recommendations followed here included not restricting respondents to a fixed number of nominations (Holland & Leinhardt, 1973; Rogers & Kincaid, 1981) and constructing the network questions to elicit typical patterns of interactions (Freeman, Romney, & Freeman, 1987).

that were present, with 0 indicating that none of an individual's contacts know each other and 1 indicating that each contact knew all the others (Marsden, 1990). Frequency was measured as the average frequency with which the managers talked to each of their contacts: "daily," "weekly," "monthly," or "less often." Component analysis of the network variables, reported in Table 1, provided support for combining these four variables into an aggregate index (factor 1), labeled range. Coefficient alpha for range is .73.

Status of contacts was defined as the average rank (superior, peer, or subordinate) of each respondent's network contacts. The component analysis results provided support for this combination: the proportion of contacts of equal and superior rank loaded on the same component (see Table 1).

Intimacy measures were derived from reports of whether respondents felt "very close," "close," "less than close," or "distant" from each contact (Burt, 1992); since the variance was largely at the extremes, I defined intimacy as the proportion of very close relationships. Although Granovetter's (1973) seminal definition of the construct of tie strength includes both frequency and intimacy, he defined tie strength as frequency in his research. Empirical evidence has since shown that frequency and intimacy are distinct dimensions of tie strength (Marsden, 1990: 455).

Multiplexity was assessed by examining respondents' responses to the five network questions in "pairwise" fashion to determine the degree of overlap between any two lists. The number of common contacts for any two network questions was multiplied by two and divided by the total (nonredundant) number of names listed for the two questions. A component analysis (Table 2) of the resulting ten overlap measures yielded three components: multiplexity of friendship ties (overlap between the social network and the four others), multiplexity of career ties (overlap between the career network and all but the friendship network), and multiplexity of task-related ties (overlap among the information, advice, and co-

TABLE 1
Results of Principal Components Analysis of Network Variables

Variables	1	2	3
Ties outside business unit	-.86	-.06	.03
Frequency	.76	.08	.21
Density	.65	.04	-.12
Ties outside company	-.59	.27	-.39
Ties to peers	.08	.85	-.11
Ties to superiors	.01	-.79	-.19
Homophily	-.02	-.05	.75
Intimacy	.09	.16	.74
Eigenvalue	2.24	1.45	1.24
Proportion of variance explained	.28	.18	.16

TABLE 2
Results of Principal Components Analysis of Multiplexity Measures

Variables	1	2	3
Overlap of friendship contracts with			
Advice	.78	.13	.12
Information	.77	-.10	.11
Cooperation	.72	.01	.02
Career	.63	.44	-.01
Overlap of career contracts with			
Information	.09	.82	.11
Advice	.14	.79	.08
Cooperation	-.17	.56	.42
Overlap among task contracts			
Advice and cooperation	.09	.18	.86
Information and cooperation	.15	-.03	.85
Information and advice	.05	.41	.59
Eigenvalue	3.15	1.86	1.26
Proportion of variance explained	.32	.19	.13

operation networks). These three components jointly explain 64 percent of the total variance, with eigenvalues of 3.15, 1.86, and 1.26, respectively. I created scales through a weighted summation of the variables loading on each component. The coefficient alphas for the three scales were .71, .67, and .74, respectively.

Homophily was measured as the proportion of same-race contacts mentioned among total citations across the five network questions.² In keeping with the focus on network strategies, I also searched the qualitative interview data for evidence of choice patterns within the minority group. Thus, an additional, qualitative indicator of homophily consisted of coded responses to the interview question, Do you seek out other minorities as network contacts? Respondent's answers fell into two discrete categories. One subgroup of the minority managers talked about actively seeking relationships with other minorities and about the importance of support networks composed of both majority and minority group members. The second subgroup reported that they did not develop minority contacts and tended to devalue or even explicitly avoid minority networks.

Thus, I created two categories for statements that reflected similarities across cases (Kram & Isabella, 1985; Mainiero, 1986): in the first type, a manager stated that minority network contact was important or useful and reported seeking out network contact with other minorities; and in the

²Such an index may be biased in that it does not take into account the total availability of minorities in the organizations (Blau, 1977). But since respondents were selected from the same hierarchical strata and organizational subgroups, similarity in the opportunity structures for whites and minorities seemed likely.

All the African Americans' minority contacts were African American. A small number of the other minority-minority dyads were not strictly homophilous (within-ethnic group).

second, the manager stated that he or she did not seek out minority network contact and did not think such contact useful. Qualitative analysis consisted simply of coding each response as falling into category 1, 2, or a default category for missing data. Each response from a minority manager was coded by me and by an independent rater on these dimensions. The initial percentage of agreement on categories was .82, and the Spearman correlation estimate of interrater reliability was .85; discrepancies across the two raters were easily resolved by discussion. Appendix B provides examples of coded quotes in each category.

The construct validity of these network measures was verified through cross-validation with the qualitative interview data, a component analysis, and a check of consistency with previous theoretical and empirical reports. Consistent with social resource theory (Campbell et al., 1986; Lin, 1982), the component analysis results reported in Table 1 provided support for distinguishing between the range and status variables. These results also supported the conceptual independence of frequency and intimacy, which loaded on different components; in the literature, researchers have viewed range as a reflection of the nonredundancy of contacts, independent of intimacy but related to frequency of contact and density (Burt, 1992; Campbell et al., 1986; Marsden, 1990).

Although the component analysis results also suggested combining homophily with intimacy, I chose not to do so for several reasons. First, the theoretical literature suggests that they should be correlated but do not indicate the same underlying construct (Marsden, 1988, 1990). Second, homophily was the variable of central theoretical interest in this research, and qualitative and quantitative sources coincided in their evidence for homophily. Research on gender differences in homophily has documented similar patterns, indicating that they are a function of organizational demography (Ibarra, 1992; McPherson & Smith-Lovin, 1987; Smith-Lovin & McPherson, 1993).

Career and task utility. On the basis of the component analysis results presented in Table 3, individuals' perceptions of the usefulness of their networks were measured as two separate variables. These were derived from

TABLE 3
Results of Principal Components Analysis of Network Utility Measures

Variables	1	2
Access to resources	.86	.01
Help learn ropes	.78	.19
Access to information	.77	.33
Help career advancement	.31	.82
Signal well-connected	-.04	.81
Access to opportunities	.29	.78
Eigenvalue	2.97	1.24
Proportion of variance explained	.50	.21

responses to questionnaire items asking respondents to rate the following on a five-point scale (1 = not at all, 5 = to a very great extent): "To what extent would you say that *your* network has been useful to you on these dimensions?" Items composing the task dimension included "helping me learn the ropes," "providing access to important information," and "providing access to resources and support." The career dimension included "gaining access to good opportunities," "as a signal that I am well connected," and "helping my career advancement." Coefficient alpha was .77 for each factor.

Race. Race was a dummy variable with a value of 1 assigned to white managers. African American, Asian American, and Hispanic managers were grouped in the minority category, which was assigned a value of 0.

Advancement potential. Since being in a fast-track or high-potential program in the four companies implied a positive judgment had been made about the managers' potential to advance to the executive ranks, I used inclusion in such a program as a proxy for opportunity for further advancement. Potential was a dummy variable with a value of 1 assigned to non-high-potential managers.

Control variables. Gender was a dummy variable with a value of 1 assigned to men. Education was also coded as a dummy variable (0 = bachelor's degree, 1 = graduate degree) and included as a control because educational level has been widely reported as having a great impact on managerial career advancement opportunities (Harlan & Weiss, 1982). Organization tenure was measured in years and included as a control because individuals' length of service with a firm can be expected to affect their informal networks. A set of three dummy variables was used to control for company-specific effects since the four firms differed in a variety of ways, including industry, performance, and organizational culture; I included these variables to ensure that network effects could be observed net of company effects not measured directly.

Data Analysis

Ordinary-least-squares regression models were the principal form of data analysis. Each regression included the control variables, gender, education, tenure, and company. To show that significant differences did not exist where they were not hypothesized to exist, I computed results for all the network characteristics considered in this investigation and present results in the tables. Support for Hypotheses 1–3 would be ascertained from significant regression coefficients for race, with the effects of advancement potential and the control variables held constant, in models with homophily, intimacy, and the three multiplexity indexes as the dependent variables. The test for Hypothesis 3 was more exploratory since the component analysis yielded three multiplexity variables.

Hypothesis 4–6 were tested by adding the interaction term crossing race by potential to the models in which homophily, range, and status were dependent variables. Following procedures for the interpretation of in-

teraction effects (Jaccard, Turrisi, & Wan, 1990), I coded the race and potential dummy variables so that the constant term in the regression model represented the estimated value (assuming values of 0 for the control variables) on the dependent variable for high-potential minorities (0 = minority, 0 = high-potential), the comparison group of theoretical interest. Estimated values for non-high-potential minorities were obtained by adding the constant to the regression coefficient for potential (0 = minority, 1 = non-high-potential); the significance of the potential coefficient constitutes the test for difference between high-potential and non-high-potential minority group members. Similarly, I obtained estimated values for high-potential whites by adding the constant to the regression coefficient for race (1 = white, 0 = high-potential); the significance of the race coefficient indicates whether high-potential whites and minorities differ on the given dependent variable.

The qualitative data were then examined for evidence supporting Hypothesis 4: Additional support would be ascertained from a finding that the high-potential minority managers were more likely than the non-high-potential minority managers to stress in their interviews the importance of developing network ties with other minorities.

In view of the component results shown in Table 3, I tested Hypotheses 7–9 in separate regression analyses for two dependent variables: career and task utility. Support for these hypotheses required significant race by range, race by status, and race by intimacy interaction terms, for Hypotheses 7, 8, and 9, respectively. The regression models were constructed hierarchically, with the control variables entered first, followed by the network variables, and finally the interaction terms. Given that one of the network characteristics, multiplexity, was measured separately for each of the two domains corresponding to the dependent variables assessing perceived network utility (task and career multiplexity were expected to correspond to task and career utility), I used only the corresponding measurement in each model.

RESULTS

Main Effects for Race

Table 4 reports variable means, standard deviations, and correlations, and Table 5 provides *t*-test results and regression coefficients for race in models for each network characteristic dependent variable. Consistent with Hypothesis 1, minority managers differed from their white peers in that the former had significantly fewer homophilous contacts. Minorities also tended to have fewer intimate relationships (Hypothesis 2), but the significance level of that relationship dropped to .10 once controls were introduced. Hypothesis 3, however, was not supported: no race-based differences were observed in any of the three multiplexity regression models. As noted below, the observed range effect disappears once the interaction of race and potential is considered.

TABLE 4
Means, Standard Deviations, and Correlations

Variables	Means	s.d.	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Gender	0.60	0.49													
2. Race ^a	0.73	0.45	-.13												
3. Potential	0.57	0.50	-.11	.05											
4. Education	0.54	0.50	-.03	-.27*	.10										
5. Tenure	14.85	7.49	.15	.20	-.06	-.28*									
6. Homophily	0.77	0.32	-.19	.90**	.14	-.32*	.11								
7. Status	2.15	0.34	.11	-.06	-.02	.03	.06	-.03							
8. Range	0.00	1.00	.01	-.31*	.22	.10	.00	-.11	.04						
9. Intimacy	0.21	0.16	-.11	.28*	-.03	-.23	.08	.22	-.12	-.18					
10. Multiplexity, social	0.00	1.00	.05	.16	.04	-.04	-.02	.08	.13	-.10	.23				
11. Multiplexity, career	0.00	1.00	.17	-.02	.00	-.20	-.17	.06	.09	.06	-.08	.21			
12. Multiplexity, task	0.00	1.00	.03	.22	-.03	-.20	.13	.25	-.02	-.15	.25*	.22	.41**		
13. Task utility	0.00	1.00	-.09	.13	.06	.08	.03	.04	-.18	-.42***	.21	.14	.00	.24	
14. Career utility	0.00	1.00	-.21	.04	.11	.11	-.14	.03	.09	-.02	-.10	.08	.20	.16	.43***

^a Race is coded as 0 = minority, 1 = white.

* $p < .05$

** $p < .01$

*** $p < .001$

TABLE 5
Differences Between Whites and Minorities in Network Characteristics^a

Dependent Variables	Means		<i>t</i>	<i>b</i>	s.e.
	Whites	Minorities			
Homophily	0.94	0.30	-15.96***	0.62***	.04
Range	-0.17	0.55	2.46*	-0.85**	.29
Status	2.14	2.17	0.28	0.02	.10
Intimacy	0.24	0.14	-2.30*	0.08†	.05
Multiplexity, social	-0.27	0.09	-1.26	0.44	.31
Multiplexity, career	0.04	-0.01	0.19	-0.01	.29
Multiplexity, task	-0.37	0.13	-1.75†	0.45	.30

^a For whites, $N = 46$; for minorities, $N = 17$. Control variables were included in the models but results are not reported.

† $p < .10$

* $p < .05$

** $p < .01$

*** $p < .001$

Interactions of Race and Potential

Table 6 reports the regression models that result from adding interaction terms crossing race by potential to the models reported in Table 5. The homophily results support Hypothesis 4. Although minorities' networks in general evidenced significantly lower levels of homophily than those of their white counterparts, the regression coefficient for potential ($b = 0.22$, $p < .01$) indicates that the minority group members with high potential for advancement had a significantly higher proportion of same-race ties than the non-high-potential minorities. The regression coefficient for race ($b = 0.52$, $p < .001$) also indicates that white high-potential managers had more homophilous ties than their minority high-potential counterparts.

These results were paralleled in the qualitative responses to the interview question, "Do you seek out other minorities as network contacts?" The majority of the high-potential minority managers reported that they sought out and valued network contact with other minorities: 75 percent expressed views consistent with those provided in the top portion of Appendix B. In contrast, a majority of the non-high-potential managers, 57 percent, responded to the same question by stating that they did not seek out minority ties; their views were consistent with the examples in the bottom portion of Appendix B.

The results in Table 6 also offer support for Hypothesis 5 and partial support for Hypothesis 6. The potential coefficients in the models having range and status as dependent variables indicate that, relative to their non-high-potential minority peers, high-potential minorities had networks that were broader in range ($b = -1.23$, $p < .05$) and lower in status ($b = 0.34$, $p < .05$). Relative to their high-potential white counterparts,

TABLE 6
Analysis of Race by Potential Interaction Effects^a

Dependent Variables	Constant		Race		Potential		Race by Potential		R ²	ΔR^2 ^b
	b	s.e.	b	s.e.	b	s.e.	b	s.e.		
Homophily	0.46***	.07	0.62***	.04	-0.06	.04			.83***	
	0.52***	.08	0.52***	.06	-0.22**	.07	0.22**	.08	.86***	.03**
Intimacy	0.21*	.08	0.08†	.06	0.00	.04			.13	
	0.20*	.09	0.09	.06	0.02	.08	-0.02	.09	.13	.00
Range	1.32**	.50	-0.85**	.29	-0.30	.25			.31**	
	1.53**	.47	-1.28***	.45	-1.23*	.52	1.18*	.57	.37**	.05*
Status	1.83***	.17	0.02	.10	0.10	.09			.16	
	1.73***	.18	0.18	.14	0.34*	.17	-0.33†	.19	.21	.05†
Multiplexity, social	0.23	.51	0.44	.31	-0.13	.27			.14	
	-0.06	.52	0.92*	.39	0.70	.52	-1.10†	.59	.20	.06†
Multiplexity, career	1.09*	.48	-0.01	.29	-0.15	.25			.24†	
	1.13*	.57	0.08	.38	-0.27	.50	0.16	.58	.24†	.00
Multiplexity, task	-0.03	.50	0.45	.29	-0.04	.26			.20	
	-0.01	.59	0.43	.39	-0.08	.52	0.05	.59	.20	.00

^a *N* = 63. Control variables are included in the models but not reported.
^b ΔR^2 indicates the change relative to the model (not reported) without the race by potential interaction term.
†*p* < .10
**p* < .05
***p* < .01
****p* < .001

however, high-potential minorities also had significantly ($b = -1.28, p < .001$) broader networks, but there was no difference in the status of the two subgroups' contacts ($b = 0.18, n.s.$). A noteworthy unhypothesized result was that high-potential minorities evidenced less social multiplexity than high-potential white managers ($b = 0.92, p < .05$).

Race and Network Utility

The final analyses explored the relationship between network characteristics and perceptions of network utility for task-related and career advancement purposes. The first model in Table 7 provides the analyses testing Hypotheses 7–9 for career utility. Hypotheses 7 and 8 were supported: Adding the interactions crossing race by range and race by status added .11 ($p < .05$) in variance explained. As Figures 1 and 2 illustrate, these significant relationships indicate that organizational range increases perceptions of networks' career utility for whites but decreases perceptions of their career utility for minority group members and that a modest increasing trend ($p < .10$) in perceived career utility is associated with having higher-ranking contacts for whites but is not present for minorities. Hypothesis 9, however, was not supported (insignificant interaction terms are not reported).

The second model in Table 7 tested Hypotheses 7–9 for task utility. Predictions were not supported: Range had similarly negative effects on perceptions of task utility for whites and minorities,³ and the interaction term (not reported) was not significant. Neither the main nor the interaction effect for status or intimacy was significant.

DISCUSSION

This study produced three sets of findings. First, when human capital variables, such as tenure and graduate education, and positional resources, such as rank and potential for advancement, were held constant, minority and white middle managers in comparable jobs differed in the homophily and intimacy of their organizational network ties. I viewed these findings as evidence of structural constraints on network patterns. Since potential for advancement was held constant, differences in ability or performance were not considered to be plausible alternative explanations for the race difference obtained. Second, evidence for the role of individual strategies, as well as support for the pluralist, rather than assimilationist, perspective was obtained from the finding that high-potential minorities differed from high-potential whites and non-high-potential minorities in the range, status, and multiplexity of their network ties. These results are consistent with the notion that strategies for managing

³The high standard errors for homophily in the models shown in Table 7 indicate multicollinearity. The results of models excluding homophily were essentially the same. The level of significance for the effects of range, however, increased to $p < .05$ in the career utility model and to $p < .01$ in the task utility model.

TABLE 7
Results of Regression Analysis^a

Variables	Dependent Variables			
	Career Utility		Task Utility	
	<i>b</i>	s.e.	<i>b</i>	s.e.
Race	−3.44*	1.70	0.46	0.73
Potential	−0.30	0.24	−0.22	0.25
Gender	−0.72**	0.26	−0.17	0.25
Education	0.12	0.26	0.23	0.26
Tenure	−0.01	0.02	0.01	0.02
Range	−0.37†	0.21	−0.34*	0.14
Status	−0.07	0.56	0.17	0.37
Homophily	−1.36	1.04	−1.23	1.07
Intimacy	0.20	0.76	0.65	0.80
Multiplexity, career	0.32*	0.13		
Multiplexity, task			0.29*	0.13
Race × range	0.48†	0.28		
Race × status	1.95*	0.77		
Constant	1.06	0.49	−0.09	1.07
<i>R</i> ²	0.43**		0.31*	
Adjusted <i>R</i> ²	0.29		0.17	
Δ <i>R</i> ^{2b}	0.11*			

^a*N* = 63. Race is coded as 0 = minority, 1 = white. Potential is coded as 0 = high-potential, 1 = non-high-potential. Company dummies were removed after a test of their contribution to model fit indicated no company effects.

^bΔ*R*² indicates change resulting from the inclusion of interaction terms.

†*p* < .10

**p* < .05

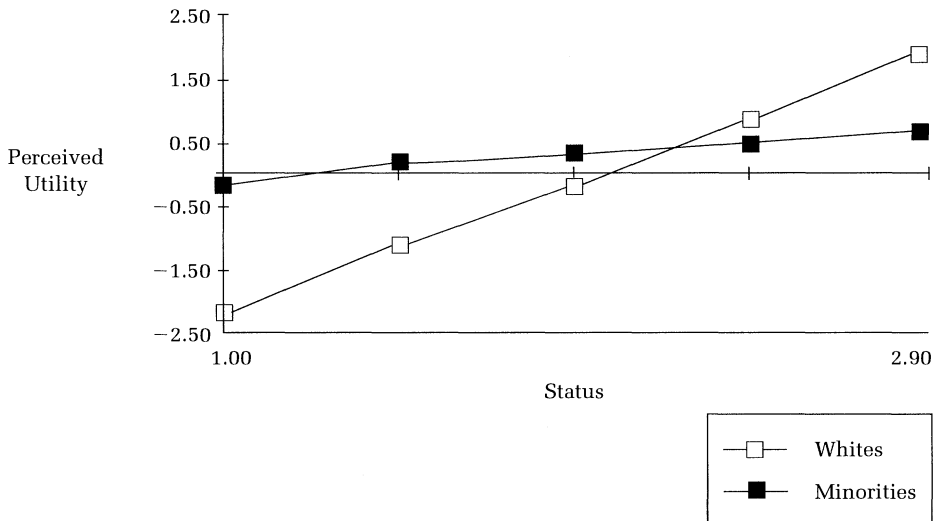
***p* < .01

structural constraints explain a variety of race differences in informal networks. Third, the race-based variation in the effects of range and status on perceptions of career utility provide additional support for a pluralist rather than an assimilationist perspective on career strategies. These effects also suggest that homophily constraints may produce unique trade-offs for minority managers that lead them to view certain network characteristics as producing fewer benefits than are produced in the view of white managers. These findings and their implications for theory and research are discussed in more detail below.

Macrostructural Constraints on Network Choices

The lower homophily found in the minority managers’ networks is consistent with the findings of a diverse body of research pertaining to organizational demography, which suggests that organizational context structures opportunities for informal interaction. To the extent that certain groups are only represented in small numbers, low availability of like others will constrain any potential preference for homophilous relationships (Blau, 1977; Kanter, 1977; McPherson & Smith-Lovin, 1987). Organiza-

FIGURE 1
Conditional Effects of Status on Perceived Career Utility of Network^a



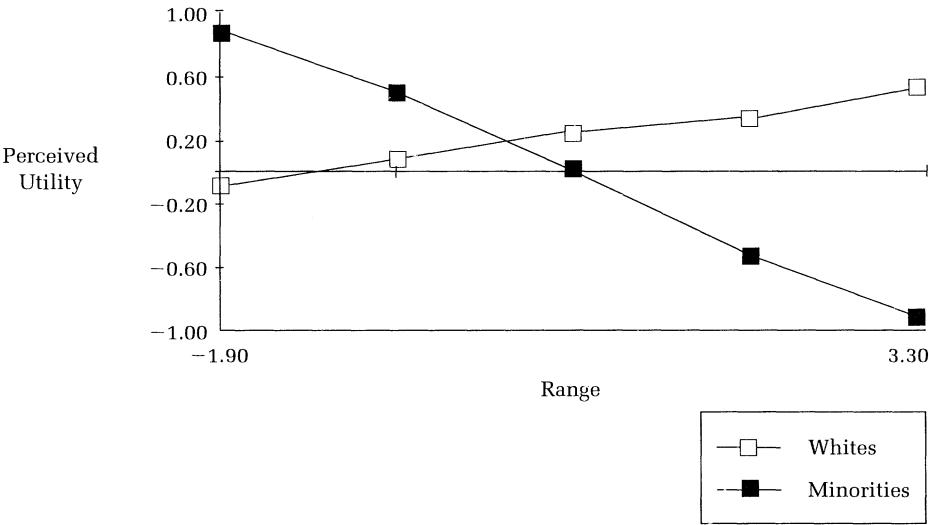
^a Estimated values of perceived career utility are based on zero values for potential, race, and gender and mean values of status, intimacy, and career multiplexity.

tional demography does not, however, affect only the homophily of a network; it may also have a direct effect on the nature of the relationships composing the network. Thus, the trend toward minority managers having fewer intimate network relationships may be understood in light of the predominance of cross-race relationships. This interpretation is consistent with the fact that homophily and intimacy loaded on the same component, as reported in the Methods section.

Although the intimacy results did not attain conventional significance levels, they are consistent with relational demography findings (Tsui & O'Reilly, 1989) that being in the minority has significant effects on individuals' affective experiences in the workplace, including isolation in work groups and lack of identification in one-on-one relationships. Extending this burgeoning literature, this trend suggests a need to examine the effects of interpersonal similarity in relationships that extend beyond immediate work groups and superior-subordinate links.

The hypothesis that minorities would differ from whites in multiplexity was not supported. My finding that only high-potential whites and high-potential minorities differed in this regard suggests that multiplexity is affected to a greater extent by network strategies than by structural constraints. With regard to the original hypothesis, however, possibly the most significant racial differences lie not in the degree of overlap between different work-related circles, but between workplace and extracurricular network circles. Overall, the relative paucity of main effects

FIGURE 2
Conditional Effects of Range on Perceived Career Utility of Network^a



^a Estimated values of perceived career utility are based on zero values for potential, race, and gender and mean values of range, intimacy, and career multiplexity.

for race in this research suggests the importance of identifying variables that may moderate the relationship between informal network structure and access to network benefits.

Network Strategies, Racial Identity, and Diverse Social Circles

The variance in network patterns observed within the minority subgroup as well as within the high-potential subgroup was viewed as evidence of the critical role of individual strategies. As Kanter (1977) noted, structure typically confronts individuals with dilemmas rather than determining outcomes. In this case, the dilemma stems from the fact that in corporate America, identity group homophily tends to be in direct competition with instrumental access for minority group members but not for whites (Ibarra, 1993). This dilemma was most clearly reflected in the racial composition of the informal networks found here, but it may also have repercussions on other aspects of network structure. Strategies for managing characteristic problems, therefore, mediate the relationship between organizational variables, such as demographic distributions, and network structure.

The network development patterns evidenced by the high-potential minority group members in this study contrast sharply with the assimilation paradigm, which suggests that to be successful, minority members must develop interpersonal networks similar to those of their white peers. Instead, these results are consistent with findings that emphasize the im-

portance of strategies for integrating racial and professional identity. More specifically, my findings parallel Thomas's (1993) finding that successful black managers develop instrumental relationships with white sponsors but do not rely on them extensively or exclusively for the psychosocial support they can obtain more readily from other minorities. In all, these results are also consistent with Cox and Nkomo's (1990b) observation that advice to black managers predicated upon the assumption of assimilation into the dominant culture may be misguided. A strategy emphasizing homophily also appears to involve bridging diverse social circles and developing networks that criss-cross not only racial boundaries but also organizational frontiers.

Speculating beyond the data suggests that the different network patterns revealed for the subgroups may reflect more complex personal and career development processes than those investigated in this study. Clearly, network strategies develop as a function of individuals' goals, identities, and circumstances (Wharton, 1992). Although I did not study racial identity directly, the interview data pertaining to network contact with other minorities suggest that an individual's dominant stance toward his or her own racial group and toward the dominant group may be a critical determinant of implicit or explicit network strategies. As noted, many of the non-high-potential minority managers did not have minority professional support networks and made no extra effort to develop them; following Thomas (1993), I would view this combination as indicating little proactive effort to integrate racial and professional identity. A potentially fruitful avenue for future research, therefore, would entail explicit measurement of racial identity and the investigation of its link to behavioral strategies, including network development approaches.

This discussion raises the question of whether differences in the minority managers' networks were the result of opportunity differentials or whether, instead, differences in network strategies and approaches contributed directly to particular individuals' being labeled high-potential. Although I have argued that having a racially diverse network may directly promote career success by helping minority managers gain access to advice and coaching from others who have had similar experiences—and may possibly have the unintended consequence of providing access to diverse sources of information—it is possible that the perception of greater opportunity may give a manager greater confidence to develop a racially diverse network. Most likely both statements are true, and future research is needed to clarify the causal order. What is clear, however, is that these results provide little support for the assimilationist perspective, which would be supported by the finding that high-potential white and minority managers had similar networks.

Network Structure and Perceptions of Network Utility

Also consistent with a pluralist rather than an assimilationist perspective on career strategies was the finding that perceptions of having a

useful network for career purposes were positively associated with range for whites but negatively related with range for minorities. Although the range by race interaction failed to reach conventional significance levels, it is consonant with Burt's (1992) finding that network range predicts career mobility for white male managers only. This trend extends Burt's theory by suggesting that homophily may be one of the mechanisms that account for the differential effects. As noted above, broad range is partially a consequence of seeking network contact with other minorities. Thus, the negative association between range and perceptions of having a useful network for career purposes may be interpreted as possibly reflecting individuals' concern about having sufficient supporters and coaches within their subunits (where promotion decisions are often made) and within the majority (white) group. Minorities may also be concerned about the signals their networks send; for example, one representative statement, reported in Appendix B, indicated concern about how others might perceive a minority network. More generally, strategies for gaining access to certain types of network resources may have unintended consequences with regard to other valued outcomes; if so, future research is needed to elucidate the nature of the results and shed light on strategies for balancing competing network requirements.

Although whites and minorities did not differ in the hierarchical composition of their networks, the effects of status on perceptions of access to career supports varied by race: it was more important for white managers' perceptions of access to career supports. This result may be interpreted in light of the small number of minority superiors available to the minority participants and the greater availability of peer mentoring for members of underrepresented groups (Kram & Isabella, 1985; Thomas, 1990); possibly, the developmental function rather than the status of the person fulfilling the function was more important to minority participants when they gauged the career utility of their networks.

To summarize the final set of results, although a variety of network characteristics were predictive of perceptions of network utility, the hypotheses concerning race-related interaction effects were only supported with regard to perceptions of utility for career-related purposes. This finding may be interpreted in light of previous reports (Davis & Watson, 1982; Kanter, 1977) suggesting that race is more likely to affect promotions, assignments, and evaluations than day-to-day interactions. Within the career domain, only two of the three hypotheses were supported; consequently, the findings discussed below should be viewed as preliminary. Perceptions of what makes for a useful network, however, are important in that they are likely to play a role in people's network development strategies. Future research is needed to investigate factors leading minority managers to see their networks as instrumentally useful.

Limitations

The most important methodological limitation of this study is that its cross-sectional design precludes statements about causal relationships. As

discussed above, the direction of the relationship between network structure and potential for advancement is difficult to determine. Further, given the apparent role of racial identity, future research is needed to explore how individual differences in developmental tasks and self-concepts shape relationship networks (Kram & Isabella, 1985). Longitudinal designs and richer qualitative data are also needed to examine differences in strategies and approaches over time.

A second limitation derives from the small size of this sample of managers and companies; a larger database will be needed to generalize this research beyond them. Although results are consistent with empirical research on minority managers in business organizations (e.g., Cox & Nkomo, 1990b; Thomas, 1990, 1993) and organizational demography (e.g., Tsui & O'Reilly, 1989; Tsui, Egan, & O'Reilly, 1992), the specific demographics of the participating firms may limit generalizability to other settings. Although the overall percentages of minority managers in these firms are equivalent to those found in corporate America today (Cox, 1991), in two of the four participating firms the percentages of African American managers exceeded the 1992 Equal Employment Opportunity Commission average of 6.5 percent (U.S. Bureau of the Census, 1993). This excess clearly raises questions concerning the effects of unmeasured company practices that may produce both the higher percentages and different networking patterns. Alternatively, whether similar patterns would be observed in organizations with larger portions of minorities in the managerial ranks is an open question.

Further, the numbers of minority women and minorities other than African Americans in this study were too small for separate analyses. The numerical dominance of African American managers clearly shaped the patterns of findings (for example, homophily tended to be lower for the Hispanic and Asian managers), and caution is advised in attempts to generalize to other minority groups. Additionally, there appeared to be a gender difference (unreported) with respect to homophily within the minority group, but the size of the subsample was too small to allow adequate exploration of this interaction effect. Future research giving more attention to multiple minority groups and to the intersection of race and gender is highly recommended (Cox & Nkomo, 1990a; Thomas & Alderfer, 1989).

The investigation of whether similar constraints and strategies distinguished between male and female managers would also be productive but was beyond the scope of this study. The social network literature suggests common constraints (Ibarra, 1993), but important differences may produce different empirical patterns. For example, in U.S. firms the numbers of female managers are currently greater than the numbers of minority managers, and the numbers of minority women in management are even fewer (Morrison & Von Glinow, 1990); thus, homophily-related effects may not be as pronounced for women. With respect to the development of intimate relationships, norms for cross-sex interaction are more fully established than norms for interracial interaction, which is characterized by

greater uncertainty (Dickens & Dickens, 1982). This effect may be mitigated, however, by uniquely gender-related issues, including sexual dynamics (Kanter, 1977; Kram, 1988). Overall, although this study may inform research on gender differences in networks, additional empirical and conceptual work is necessary before these conclusions can be so generalized.

Few network studies including significant numbers of minority group members in managerial positions have been conducted to date, and most studies of the effects of race on organizational experiences have either lacked theoretical underpinnings (Cox & Nkomo, 1990a) or been conducted within a narrowly defined paradigm that fails to cast race as an integral part of a "complicated web of socially constructed elements of identity formation" (Nkomo, 1992: 507). The alternative perspective advocated here confirms the pervasiveness of structural constraints but also emphasizes the role of strategic action and indicates a critical role for social identity. Although the present results are only a starting point, research on the unique opportunities and constraints minority managers face in developing useful networks of organizational relationships may prove a promising area for future inquiry.

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APPENDIX A

Interview Protocol

1. Tell me about a few significant relationships in your network.
2. Give me an example of a situation where having a relationship with one or a subset of the people in your network proved to be critical to your ability to get something done that was important to you.

3. How would you describe your approach to building and developing your network of relationships? What's important to you in a network relationship? What kinds of people do you try to develop relationships with?
4. What are principal advantages and disadvantages of having this network of relationships? What do like best about it/what works well? What would you change/improve?
5. Do you seek out network contact with other minority employees? Why or why not?

APPENDIX B

Examples of Coded Quotes

Category 1: Values and Seeks Minority Contact

With the minority group you get the camaraderie and the strength that allows you to go back and do your job effectively. You can be yourself—we've all had the same struggles. You don't need any cues, and you don't have the heightened sensitivity to every act. In the white circle, there is a script and a director and a producer who have a vision about what is to be done. Keeping the two circles separate allows you to be successful, to have that mental clarity rather than having an internal battle.

I try to get to know the other black people in the organization. I steer towards those with similar positions or higher because I'm looking to move up. They can alert me to the political issues in the organization. . . . [Name] is the highest ranking Black manager at [firm]. I go to him to find out what is going on, to ask how I'm doing and what he hears about me. . . .

Category 2: Does Not Value and Seek Minority Contact

In [company] there is a definite network of black employees. But it's more a support group because we are underrepresented, we're not going anywhere in the corporation. My contacts are all based on whether we have business together. I haven't done anything to get into the black network and I haven't found the time to deviate from the business task. They don't have any business connection but still spend a lot of time together. Maybe I have consciously tried to stay away from them. We're very visible when we congregate. People wonder what we're plotting. . . . All of my relationships are work-related. When I'm at home, I'm somebody. When I'm at work I'm a [title] not [name], the individual.

[Networking with blacks] gives you a place to sound off and express yourself but I'm not sure that it has been useful for anything other than venting frustrations and relieving stress. . . . You will find that those black managers who have been successful probably don't interact much socially with other blacks.

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